

Math 520, F15

Name: Answer

Quiz 13

SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Find the solution of the given initial value problem:

$$x' = \begin{pmatrix} -3 & 2 \\ -1 & -1 \end{pmatrix} x, \quad x(0) = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

Solution: $A = \begin{pmatrix} -3 & 2 \\ -1 & -1 \end{pmatrix}$

$$\det(A - \gamma I) = \det \begin{pmatrix} -3-\gamma & 2 \\ -1 & -1-\gamma \end{pmatrix} = (-3-\gamma)(-1-\gamma) + 2$$

$$= \gamma^2 + 4\gamma + 5$$

$$\Rightarrow \gamma_1 = -2+i, \quad \gamma_2 = -2-i$$

Solve $(A - \gamma_i I) \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = 0$, we get

$$\begin{pmatrix} -1-i & 2 \\ -1 & 1-i \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = 0$$

$$-\xi_1 + (1-i)\xi_2 = 0$$

$$\Rightarrow \xi_1 = 1-i, \quad \xi_2 = 1 \Rightarrow \xi^{(1)} = \begin{pmatrix} 1-i \\ 1 \end{pmatrix}$$

$$\Rightarrow x^{(1)}(t) = e^{\gamma_1 t} \xi^{(1)} = e^{(-2+i)t} \begin{pmatrix} 1-i \\ 1 \end{pmatrix}$$

$$= e^{-2t} (\cos t + i \sin t) \begin{pmatrix} 1-i \\ 1 \end{pmatrix} = e^{-2t} \begin{pmatrix} \cos t + \sin t + i(\sin t - \cos t) \\ \cos t + i \sin t \end{pmatrix}$$

$$= e^{-2t} \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix} + i e^{-2t} \begin{pmatrix} \sin t - \cos t \\ \sin t \end{pmatrix}$$

$$\Rightarrow u = e^{-2t} \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix}, \quad v = e^{-2t} \begin{pmatrix} \sin t - \cos t \\ \sin t \end{pmatrix}$$

$$\Rightarrow x(t) = C_1 u + C_2 v = C_1 e^{-2t} \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix} + C_2 e^{-2t} \begin{pmatrix} \sin t - \cos t \\ \sin t \end{pmatrix}$$

$$x(0) = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \Rightarrow C_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \Rightarrow \begin{matrix} C_1 - C_2 = -2 \\ C_1 = 3 \end{matrix}$$

$$\Rightarrow C_1 = 3, \quad C_2 = 5$$

Thus $x(t) = e^{-2t} \begin{pmatrix} -2\cos t + 8\sin t \\ 3\cos t + 5\sin t \end{pmatrix}$